Identification of Gases

Exam code: 0620 & 0971

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Updated on 19 January 2026

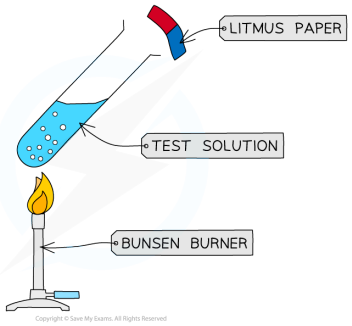
Identification of gases

- Several tests for anions and cations produce **gases** which then need to be tested
- The gases included in the syllabus are:
  - Ammonia
  - Carbon dioxide
  - Chlorine
  - Hydrogen
  - Oxygen
  - Sulfur dioxide

Test for ammonia

- Ammonia is a gas with a characteristic sharp choking smell that turns damp **red** litmus paper **blue**
- Hold the litmus paper near the mouth of the test tube, but be careful to avoid touching the sides of the test tube
- If you are testing for ammonia produced from ammonium ions and sodium hydroxide, avoiding touching the sides to prevent traces of sodium hydroxide from also turning the red litmus paper blue

Testing for ammonia gas



Damp red litmus paper turns blue in the presence of ammonia

Examiner Tips and Tricks

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

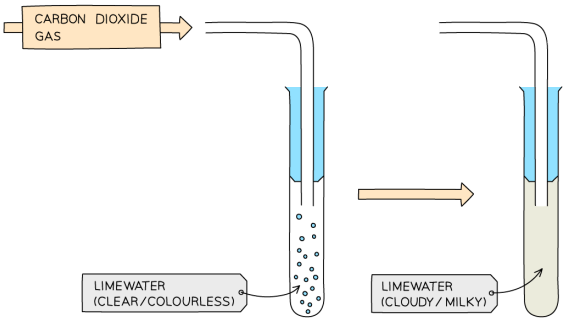
Got it

- Ammonium refers to the aqueous cation, NH_4^+
- Ammonia refers to the gas, NH_3.

Test for carbon dioxide

- The test for carbon dioxide involves bubbling the gas through an aqueous solution of **limewater** (calcium hydroxide)
- If the gas is carbon dioxide, the limewater turns **cloudy white**

Testing for carbon dioxide gas



Limewater turns cloudy white in the presence of carbon dioxide

Examiner Tips and Tricks

Sometimes students think that extinguishing a burning splint indicates carbon dioxide gas.

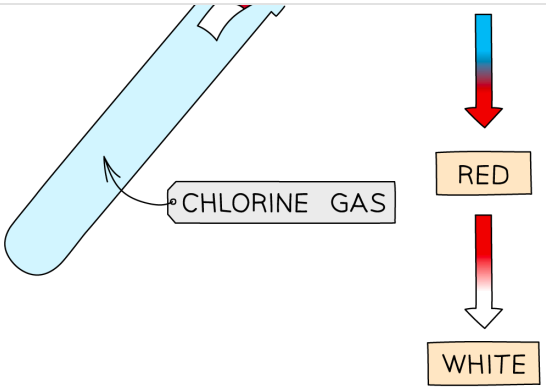
However, while it is a property of carbon dioxide, other gases, such as nitrogen, will also do this.

So, the test is not definitive and should not be given as an exam answer.

Test for chlorine gas

- The test for chlorine makes use of **litmus paper**
- If chlorine gas is present, damp blue litmus paper will turn red and then be **bleached white**
  - It turns red initially as acids are produced when chlorine comes into contact with water
- Chlorine also has a characteristic sharp, choking smell
- Chlorine should always be handled in a fume cupboard due to its toxicity

Testing for chlorine gas



Chlorine bleaches damp blue litmus paper white

Examiner Tips and Tricks

You should distinguish between properties of gases and tests for gases. Chlorine 'smells like swimming pools' is a characteristic, but it is not an acceptable means of identification.

Test for hydrogen gas

- The test for hydrogen consists of holding a **burning splint** at the open end of a test tube of gas
- If the gas is hydrogen it burns with a loud "**squeaky pop**" which is the result of the rapid combustion of hydrogen with oxygen to produce water
- Be sure not to insert the splint right into the tube, just at the mouth, as the gas needs air to burn

Testing for hydrogen gas

Examiner Tips and Tricks

It is easy to confuse the tests for hydrogen and oxygen.

Try to remember that a lig**H**ted splint has an **H** for Hydrogen, while a gl**O**wing splint has an **O** for Oxygen.

Test for oxygen

- The test for oxygen consists of placing a **glowing splint** inside a test tube of gas
- If the gas is oxygen, the splint will **relight**

Testing for oxygen gas

A glowing splint will relight in the presence of oxygen

Examiner Tips and Tricks

Sometimes the splint does not relight, but it glows very brightly, which is also a positive result. In an exam, however, it is best to state it relights the glowing splint.

Test for sulfur dioxide

- Sulfur dioxide is a gas with a characteristic sharp choking smell
- The test for sulfur dioxide involves bubbling the gas through an acidified solution of **potassium manganate(VII)**

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